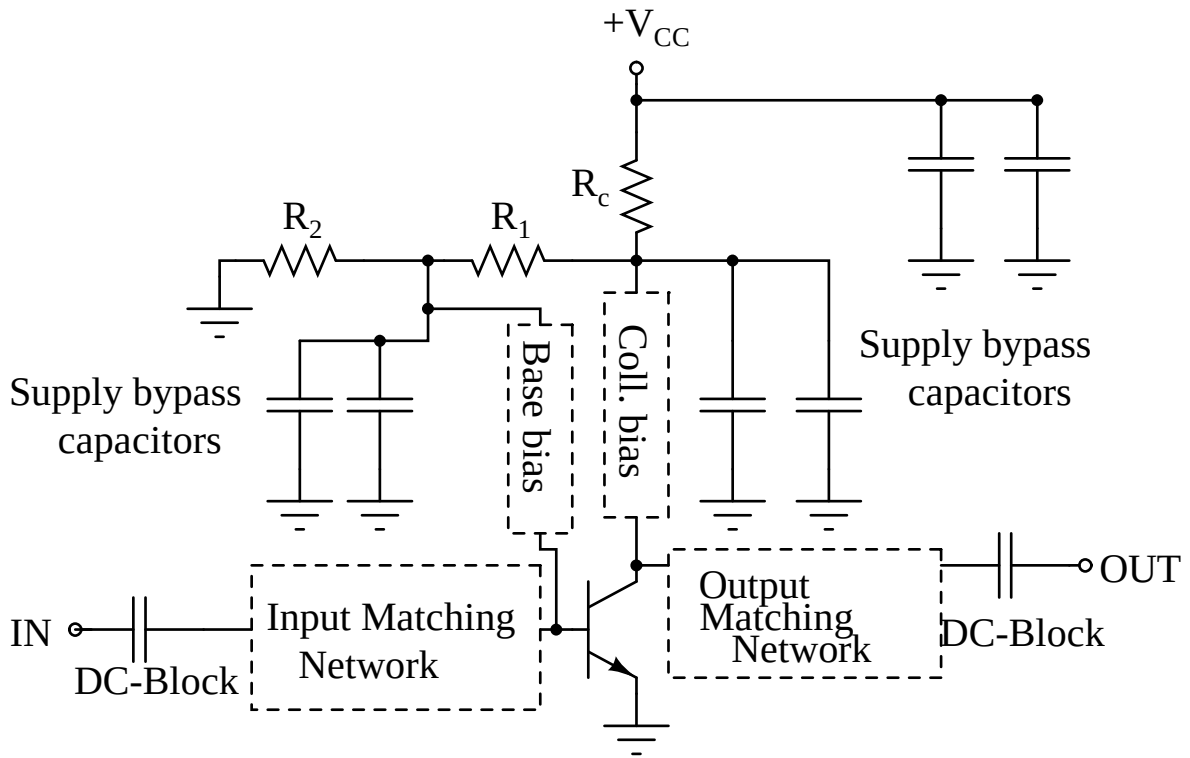


Hints for the project

You need to bias the transistor properly using a β -independent biasing circuit. A recommended circuit follows:



Choose the operating point (V_{CE} , I_C)

The collector current can be found from the following equation:

$$I_C = \beta \{ V_{CC} - V_0 [1 + (R_c + R_1)/R_2] \} / [(\beta + 1)R_c + R_1]$$

For good β stability, it is a good idea to set $V_0/R_2 = 10 I_B$

The collector to emitter voltage can be found from

$$V_{CE} = V_{CC} - [I_C(\beta + 1)/\beta + V_0/R_2] R_c$$

DC block capacitors should be selected to have a reactance of 1 to 5 Ω at the operating frequency.

The supply bypass capacitors should be selected to give 1 to 5 Ω range at 500 kHz and 50 MHz.